

A Hypergraph-Native Message Passing Layer Based on Eidi-Otter Connectivity Patterns

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Abstract

This note addresses the question of whether one can define a message passing paradigm for hypergraphs directly in terms of hyperedge connectivity patterns, rather than by first reducing the hypergraph to a graph-based or relational representation. Motivated by the constructions of Eidi and Otter [2], we propose a vertex-centered message passing layer whose propagation rule is built from the equal-edges random walk and from a cardinality-pattern descriptor associated with the hyperedges shared by two vertices. The resulting architecture retains the standard message/aggregate/update structure of an MPNN [3], but the interaction between two vertices is not determined only by their adjacency in the induced pairwise graph. Instead, messages are weighted by a hypergraph-native transition kernel and conditioned on the cardinalities of the hyperedges through which the vertices interact.

The purpose of the construction is to isolate a minimal hypergraph-native paradigm that can be compared conceptually with clique-expansion methods and relational message passing frameworks. In particular, the proposed layer does not start from an influence graph or from a general relational structure, but from the incidence and cardinality data of the hypergraph itself. We show that the layer is permutation equivariant, discuss its computational cost, and identify possible extensions such as curvature-based gates. This provides a first concrete candidate suggesting that hypergraph message passing can be formulated directly from hyperedge connectivity patterns, at least in a vertex-centered and cardinality-sensitive form. Further empirical and theoretical validation is needed to assess its expressive and practical advantages.

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1 Eidi-Otter EE-Pattern Message Passing

The aim of this short report is to propose a simple hypergraph-native message passing paradigm inspired by the constructions of Eidi and Otter [2]. The guiding principle is that message passing on a hypergraph should not be defined only by the pairwise adjacency induced by its hyperedges, but by the hyperedge connectivity patterns through which two vertices interact.

Standard Message Passing Neural Networks (MPNNs) update node representations by aggregating messages from neighbouring nodes [3]. In the graph case, the neighbourhood relation is given by edges. In the hypergraph case, however, a hyperedge may contain more than two vertices, and reducing it to pairwise edges may erase part of the higher-order information. This is precisely the distinction that motivates hypergraph neural networks such as AllSet [1], where propagation is performed through node-to-hyperedge and hyperedge-to-node multiset functions.

Our goal here is slightly different. We want a vertex-centered MPNN whose propagation operator is defined directly from the hypergraph geometry introduced by Eidi and Otter [2]. In particular, we use their equal-edges random walk as the basic propagation kernel and augment each message with a structural descriptor encoding the cardinalities of the hyperedges through which the source and target vertices are connected.

This contrasts with the relational-structure viewpoint of Taha et al. [4], where topological message passing is formulated through a finite set of entities, relations, shift operators, and an associated influence graph. That framework is useful as an axiomatic and analytical language, especially for studying oversquashing in topological message passing. However, the construction proposed here does not start from a relational graph or an influence graph. Instead, the message passing rule is defined directly on the hypergraph by using hyperedge incidence and cardinality-sensitive connectivity.

1.1 Hypergraph notation

Let

$$H = (V, \mathcal{E})$$

be a finite undirected hypergraph, where V is the set of vertices and $\mathcal{E} \subseteq 2^V \setminus \{\emptyset\}$ is the set of hyperedges. For a hyperedge $e \in \mathcal{E}$, its cardinality is denoted by $|e|$.

For a vertex $v \in V$, define the set of incident hyperedges of size at least two by

$$\mathcal{E}(v) := \{e \in \mathcal{E} : v \in e, |e| \geq 2\}.$$

The hypergraph degree of v is

$$d_H(v) := |\mathcal{E}(v)|.$$

For two distinct vertices $v, u \in V$, define the set of hyperedges jointly incident to v and u by

$$\mathcal{E}(v, u) := \{e \in \mathcal{E} : v \in e, u \in e\}.$$

The hypergraph neighbourhood of v is then

$$N_H(v) := \{u \in V \setminus \{v\} : \mathcal{E}(v, u) \neq \emptyset\}.$$

Thus, $u \in N_H(v)$ whenever u and v co-occur in at least one hyperedge. However, unlike a clique expansion, we do not forget which hyperedges generate this adjacency. The set $\mathcal{E}(v, u)$ remains part of the message passing rule.

1.2 The Eidi-Otter equal-edges kernel

Eidi and Otter define several hypergraph-sensitive notions of similarity and equivalence between vertices. A key distinction is between weak notions, which only remember the existence of shared neighbours, and stronger notions, which also remember the cardinalities of the hyperedges through which those neighbours are reached [2].

The most useful primitive for our purposes is the equal-edges random walk. This random walk defines the probability of moving from a vertex v to another vertex u in one step. Starting at v , the walker first chooses uniformly one hyperedge incident to v . Since there are $d_H(v)$ such hyperedges, each one is chosen with probability $1/d_H(v)$.

If the chosen hyperedge is e , the walker then chooses uniformly one of the $|e| - 1$ vertices of e different from v . Therefore, each hyperedge e containing both v and u contributes a term

$$\frac{1}{d_H(v)} \frac{1}{|e| - 1}$$

to the transition probability from v to u . Summing over all hyperedges containing both vertices gives

$$P_{EE}(v, u) = \frac{1}{d_H(v)} \sum_{e \in E(v, u)} \frac{1}{|e| - 1}, \quad u \neq v,$$

whenever $d_H(v) > 0$. If $d_H(v) = 0$ ¹, one may set $P_{EE}(v, v) = 1$ and $P_{EE}(v, u) = 0$ for $u \neq v$.

The factor $1/(|e| - 1)$ is essential. It means that a neighbour reached through a small hyperedge receives more weight than a neighbour reached through a very large hyperedge. Therefore, P_{EE} is not simply the random walk on the clique expansion. It is a cardinality-sensitive hypergraph propagation operator.

1.3 Cardinality-pattern summary

The transition kernel $P_{EE}(v, u)$ determines how much weight vertex v assigns to the neighbour u during aggregation. However, this scalar weight does not fully describe the way in which v and u are connected. As emphasised by Eidi and Otter [2], hypergraph connectivity can be considered at different levels of resolution: weak notions remember only the existence of shared neighbours, whereas stronger notions also take into account the cardinalities of the hyperedges through which those neighbours are reached.

Motivated by this distinction, we augment the equal-edges transition weight with a learnable cardinality-sensitive summary. Two vertices may be connected through a small hyperedge, through a large hyperedge, or through several hyperedges of different cardinalities. These cases should not necessarily produce the same type of message.

To encode this information in a learnable way, let $a_r \in \mathbb{R}^p$ be a learnable embedding associated with hyperedge cardinality $r \geq 2$. Intuitively, a_r represents the message-passing type associated with an interaction mediated by a hyperedge of size r .

For each pair v, u with $E(v, u) \neq \emptyset$, define the normalising factor

$$Z_{vu} := \sum_{e \in E(v, u)} \frac{1}{|e| - 1}.$$

¹If $d_H(v) = 0$, then v is isolated from the point of view of the equal-edges walk: it is not incident to any hyperedge of size at least two.

We then define the cardinality-pattern descriptor

$$\chi_{vu} := \frac{1}{Z_{vu}} \sum_{e \in E(v,u)} \frac{1}{|e| - 1} a_{|e|}.$$

Thus, χ_{vu} is a weighted average of the learnable embeddings corresponding to the cardinalities of the hyperedges shared by v and u . The weighting is inherited from the equal-edges kernel: interactions mediated by smaller hyperedges contribute more strongly, while larger hyperedges distribute their influence among more possible neighbours.

The role of χ_{vu} is therefore to refine pairwise adjacency with cardinality information from the original hypergraph. It does not encode the full multiset of shared hyperedges, and different local incidence patterns may lead to the same descriptor. Rather, it provides a cardinality-sensitive summary that allows the message function to distinguish some hypergraphs that would be identical after an unweighted clique expansion.

For example, consider two hypergraphs on vertices $\{a, b, c\}$:

$$H_1 : \mathcal{E}_1 = \{\{a, b, c\}\}, \quad H_2 : \mathcal{E}_2 = \{\{a, b\}, \{a, c\}, \{b, c\}\}.$$

Both induce the same clique expansion, namely a triangle. However, for the pair (a, b) , the first hypergraph gives a cardinality pattern corresponding to a hyperedge of size 3, while the second gives a cardinality pattern corresponding to a hyperedge of size 2. Hence

$$\chi_{ab}^{H_1} = a_3, \quad \chi_{ab}^{H_2} = a_2.$$

Thus, the proposed layer can distinguish these two hypergraphs despite their identical unweighted clique projection. This illustrates that χ_{vu} acts as a cardinality-sensitive refinement of pairwise adjacency, although not as a complete invariant of higher-order incidence structure.

1.4 The EO-Pattern MPNN layer

Each vertex $v \in V$ carries a hidden representation

$$h_v^{(\ell)} \in \mathbb{R}^{d_\ell}$$

at layer ℓ . The goal of one message passing layer is to update this representation by aggregating information from the hypergraph neighbours of v .

For each neighbour $u \in N_H(v)$, we construct a message using three pieces of information: the current representation of the target vertex v , the current representation of the source vertex u , and the cardinality-sensitive descriptor χ_{vu} summarising the sizes of the hyperedges through which v and u are jointly incident. This gives a learnable message of the form

$$\psi^{(\ell)} \left(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu} \right),$$

where $\psi^{(\ell)}$ may be implemented, for instance, as an MLP.

The message from u to the update of v is then weighted by the equal-edges transition probability $P_{EE}(v, u)$. Thus, the aggregated message at v is

$$m_v^{(\ell+1)} = \sum_{u \in N_H(v)} P_{EE}(v, u) \psi^{(\ell)} \left(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu} \right).$$

Finally, the new representation of v is obtained by applying a learnable update function $\phi^{(\ell)}$ to the previous representation of v and to the aggregated message:

$$h_v^{(\ell+1)} = \phi^{(\ell)} \left(h_v^{(\ell)}, m_v^{(\ell+1)} \right).$$

A simple concrete choice is

$$h_v^{(\ell+1)} = \text{MLP}^{(\ell)} \left(h_v^{(\ell)} \parallel m_v^{(\ell+1)} \right),$$

where $\parallel$ denotes concatenation.

This architecture has the standard message/aggregate/update structure of an MPNN [3]. However, the interaction between v and u is not determined only by their adjacency in the induced pairwise graph. The scalar weight $P_{\text{EE}}(v, u)$ is obtained from the equal-edges hypergraph random walk, while the message function is conditioned on the cardinality-pattern descriptor χ_{vu} . Hence, the layer uses the induced neighbourhood as the support of aggregation, but the propagation rule itself remains sensitive to the hypergraph structure that generated that neighbourhood.

1.5 Why this is hypergraph-native

The proposed layer is hypergraph-native in the following sense. Its propagation rule is not obtained by first replacing the hypergraph with a graph and then applying a graph neural network. Instead, the layer uses two quantities defined intrinsically from the hypergraph:

1. The equal-edges transition weight $P_{\text{EE}}(v, u)$.
2. The cardinality-pattern descriptor χ_{vu} .

The first quantity controls the strength of propagation from u to v . The second controls the structural type of the message. Together, they encode not only that two vertices are adjacent through the hypergraph, but also a cardinality-sensitive summary of the incident hyperedges that generate this adjacency. Therefore, the construction is hypergraph-native in a precise but limited sense: the support of aggregation is still pairwise and vertex-centered, but the weights and message descriptors are computed directly from the original hypergraph incidence structure rather than from an unweighted graph projection.

This is the key difference with clique expansion. A clique expansion keeps only the fact that two vertices co-occur in at least one hyperedge. The proposed layer also keeps information about the cardinalities of those hyperedges. Therefore, it is sensitive to higher-order connectivity patterns that pairwise graph projections may erase.

1.6 Relation to relational message passing

The construction should also be distinguished from the relational-structure framework of Taha et al. [4]. In that framework, a higher-order object is represented as a relational structure

$$R = (S, R_1, \dots, R_k),$$

with shift operators associated with the relations. These shift operators are aggregated into influence matrices, which in turn define an influence graph: a directed weighted graph on the entity set S ,

with an edge $\tau \rightarrow \sigma$ whenever the aggregated influence of τ on σ is non-zero. This influence graph is then used to analyze information propagation and oversquashing.

The present construction does not use a relational structure as its starting point. It starts directly from the hypergraph incidence pattern. The message passing rule is determined by $\mathcal{E}(v, u)$, $|e|$, P_{EE} , and χ_{vu} . Of course, as with many finite message passing schemes, one could later encode this model inside a sufficiently rich relational framework. But this would be an after-the-fact representation, not the conceptual origin of the architecture.

Thus, the distinction is not that the proposed layer is impossible to relationalize. The distinction is that the primitive object defining propagation is the hypergraph connectivity pattern itself.

1.7 Permutation equivariance

The proposed layer is permutation equivariant with respect to relabellings of the vertex set.

Let $\pi : V \rightarrow V$ be a permutation of the vertices, and let πH be the relabelled hypergraph with hyperedges

$$\pi \mathcal{E} := \{\pi(e) : e \in \mathcal{E}\}, \quad \pi(e) := \{\pi(v) : v \in e\}.$$

Then

$$\mathcal{E}_{\pi H}(\pi v, \pi u) = \{\pi(e) : e \in \mathcal{E}_H(v, u)\}.$$

Moreover, $|\pi(e)| = |e|$ and $d_{\pi H}(\pi v) = d_H(v)$. Hence

$$P_{EE}^{\pi H}(\pi v, \pi u) = P_{EE}^H(v, u),$$

and

$$\chi_{\pi v, \pi u}^{\pi H} = \chi_{vu}^H.$$

Since the message function $\psi^{(\ell)}$ and the update function $\phi^{(\ell)}$ are shared across all vertices, and since aggregation is performed by summation over an unordered neighbourhood, the resulting layer satisfies

$$h_{\pi v}^{(\ell+1)}(\pi H) = h_v^{(\ell+1)}(H)$$

after relabelling the hidden states accordingly. Therefore, the layer is vertex permutation equivariant.

1.8 Computational cost

A direct implementation of the layer can be performed by iterating over hyperedges. For each hyperedge e , and for each vertex $v \in e$, one has to account for the possible messages coming from every other vertex $u \in e$, with $u \neq v$. Thus, a hyperedge of size $|e|$ contributes

$$|e|(|e| - 1)$$

ordered pairwise interactions.

Therefore, the total cost of constructing the pairwise propagation data is

$$O\left(\sum_{e \in E} |e|^2\right).$$

This is the same worst-case pairwise cost as constructing a clique expansion, since a large hyperedge also induces quadratically many pairwise interactions. However, the proposed layer does not semantically collapse the hypergraph to its clique expansion. The pairwise interactions are used only as the support of message passing, while each interaction still carries hypergraph-specific information through the equal-edges weight $P_{\text{EE}}(v, u)$ and the cardinality-pattern descriptor χ_{vu} .

The quadratic dependence on $|e|$ may become expensive for datasets containing very large hyperedges. In that case, one can approximate the aggregation by sampling a subset of vertices inside each hyperedge, analogously to neighbourhood sampling in large-scale graph neural networks. This reduces the cost, at the price of replacing the exact aggregation by a stochastic approximation.

1.9 Possible extensions

The base layer deliberately avoids additional mechanisms such as hyperedge hidden states, hyperedge-to-hyperedge propagation, quotient-level summaries, or curvature gates. These extensions may be useful, but they are not necessary to define the core hypergraph-native paradigm.

A natural extension inspired by Eidi and Otter [2] would be to include an Ollivier–Ricci curvature gate. If $\kappa_{\text{EE}}(v, u)$ denotes the equal-edges Ollivier–Ricci curvature associated with the pair (v, u) , one could replace the propagation weight by

$$\tilde{P}_{\text{EE}}(v, u) = P_{\text{EE}}(v, u) g(\kappa_{\text{EE}}(v, u)),$$

where g is either a fixed scalar function or a learnable gate. This would make the layer sensitive not only to local hyperedge cardinalities, but also to the local geometry induced by optimal transport. Nevertheless, this curvature-gated version should be considered an extension, not the minimal model. The goal of this first proposal is to isolate the minimal hypergraph-native message passing paradigm and to check whether it is mathematically well-defined and implementable. More sophisticated mechanisms can then be studied once the core model has been validated, both theoretically and empirically.

1.10 Summary

The proposed Eidi–Otter EE-Pattern MPNN is the layer

$$m_v^{(\ell+1)} = \sum_{u \in N_H(v)} P_{\text{EE}}(v, u) \psi^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}),$$

$$h_v^{(\ell+1)} = \phi^{(\ell)}(h_v^{(\ell)}, m_v^{(\ell+1)}),$$

where

$$P_{\text{EE}}(v, u) = \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1},$$

and

$$\chi_{vu} = \frac{1}{Z_{vu}} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1} a_{|e|}.$$

This gives a simple, vertex-centered, permutation-equivariant message passing paradigm for hypergraphs. Its main feature is that messages are not only weighted by hypergraph incidence, but are

also conditioned on the cardinality pattern of the hyperedges through which vertices communicate. In this precise sense, the model is based on hypergraph-derived connectivity information rather than on an unweighted clique projection. More precisely, it remains a vertex-centered pairwise aggregation scheme, but the pairwise weights and message descriptors are induced by the original hyperedge incidence and cardinality structure.

References

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